Rank-27 Post-Trail Propagation Closures

Tactix Development Project

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Executive summary

The accepted reserved assignment-trail repair remains the latest solver change at synchronized commit `21aa151bdaf6e881ece0a273ff918f6bf4693465`. A fresh exact-pin full profile, job `1645779`, completes the same checked Rank-27 trajectory in 403.887 internal seconds. Ordinary watched propagation remains the leading owner at approximately 195.780 sampled cycles per printed propagation, versus 174.239 for the already accepted current-source Kissat diagnostic. Frozen Kissat 4.0.4 was not rerun, and current Kissat remains diagnostic-only.

Six source-backed Tactix experiments were therefore priced simultaneously with the accepted predecessor at the exact conflict-9450 common state. Every arm preserves all printed non-time counters, rule work, logical ticks and costs, candidate and scheduling order, learning, and pruning. None improves both complete elapsed time and complete sampled cycles materially. All six are rejected and fully reverted. One experiment did reduce its intended propagation owner by 1.22%, but regressed the complete prefix; it is not a promotion candidate.

The first atomic large-watch prototype also exposed a reproducible heap corruption before submission. The repository-mandated bounded Claude workflow repaired and regression-tested the exact-capacity defect, and independent Release plus ASan/UBSan checks passed. The repaired candidate was then measured as neutral and reverted. No crashing binary reached SLURM, and every campaign has zero certificate or staged-input residue. The next convergence boundary is the per-list watch-head/flat-arena addressing and relocation population, not a reopening of the six closed assignment/clause micro-hypotheses.

1 Authority and full-run localization

The immutable Rank-27 input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. All Rank-27 jobs use two CPUs. The frozen checked Rank-27 target remains 300 seconds; the accepted reserved-trail witness, job `1645778`, is valid UNSAT in 407.006 internal seconds with its byte-identical 1,514,395,111-byte DRAT proof independently verified and deleted.

Campaign `benchmarks/runs/20260824-1520-tierd-rank27-reserved-trail-full-profile/`, job `1645779`, profiles the exact accepted binary SHA-256 `b6c1623b89477b140cabddb3cb139ca797778a871aeb67ab35fc708c3cd87851`. It returns the expected UNSAT verdict in 403.887 internal / 404.57 profiled wall seconds with 3,813,669 conflicts, 175,272,633 decisions, and 2,512,356,477 propagations. The cycles profile records 80,464 samples, approximately 1.51993 trillion cycles, and zero lost samples. Ordinary full propagation owns 32.36%, or 491,868,070,611 sampled cycles. Normalized by the printed propagation count this is approximately 195.780 cycles per propagation; the accepted current-Kissat task from job `1645700` is 174.239. The remaining 1.12362 ratio motivates

exact-state representation tests, but is not itself behavioral repair authority because the complete solver trajectories differ.

2 Exact common-state dispositions

Each campaign below uses the same immutable input, predecessor SHA `68a2ae6bb444fd54610485 41bff366926a19f3c642815a6fe65d212091e5b9d1`, two CPUs, 16 GiB, a 300-second task timeout, and a conflict-9450 stop. Both arms always report 9,450 conflicts, 5,795,248 decisions, and 96,338,812 propagations. No proof is requested for these diagnostic profiles; both staged CNFs are removed on every exit.

Job	Representation hypothesis	Complete result	Disposition
1645781	cold clause proof ID	cycles 1.01933x	reject
1645783	active-only assignment	owner 0.99913x	reject
1645785	binary-source snapshot	owner 1.00026x	reject
1645787	saved-phase snapshot	owner 1.00927x	reject
1645789	no-ignore scan	total 1.00321x	reject
1645791	atomic large-watch push	total 0.99882x	neutral/reject

Clause and assignment representation. Campaign `benchmarks/runs/20260824-1535-tie rd-rank27-cold-clause-id-profile/` moves proof-ID words behind semantic literals, making Tactix's watched literal offsets match Kissat's 12/16-byte positions. Propagation cycles improve only 19.023B to 18.835B while total cycles rise 64.298B to 65.541B and elapsed time rises 16.763 to 17.034 seconds. The active-only assignment arm (`1645783`) removes a retired-variable test only from DefaultPolicy search, preserving N1's general retired-variable decision surface. Its intended owner is unchanged at 0.99913x. Hoisting the immutable binary source assignment (`1645785`) is likewise flat at 1.00026x propagation cycles; the compiler already retained that metadata effectively.

Control and relocation representation. The saved-phase snapshot arm (`1645787`) resolves ownership and suppression once per scan, but the prepared-call shape raises propagation cycles 1.00927x and complete cycles 1.01003x. The no-ignore arm (`1645789`) follows Kissat's compile-time separation between ordinary and probing propagation. It does move the intended owner: combined propagation cycles fall 19.011B to 18.778B, or 0.98777x. Complete elapsed time nevertheless rises 16.799 to 16.853 seconds and complete cycles rise 1.00321x. A local improvement that does not improve the exact whole prefix does not justify duplicated hot code.

Finally, job `1645791` appends a delayed two-word large watch after one common-path capacity decision while retaining exact blocking-word/reference order. The repaired form is neutral: 16.820 versus 16.822 seconds, 64.518B versus 64.442B total cycles, and 18.788B versus 18.767B propagation cycles. The 0.1% changes are below materiality and the simpler two-push code is restored.

3 Crash escalation and independent validation

The initial atomic-watch prototype assumed that one doubling always leaves two free words. Exact batch rebuilds and defragmentation can instead leave `capacity == size == 1`; one growth then produces capacity two and only one free word. The prototype wrote past the arena and made `solve_reset` segfault after its matching reused/fresh solve summaries.

This reproduced crash triggered DEVELOPMENT_PLAN §8.2's mandatory bounded Claude escalation. Claude was granted workspace edit permission and restricted to the two-word append, its reproducer, and focused checks. The repair loops until two words actually fit, retains one common-path capacity decision, and adds an exact one-word-slice regression. Claude reports all 73 code tests passing; only the pre-existing opt-in documentation-size check fails.

Codex independently reviewed the patch and reran Release `kernel` plus `solve_reset`; both pass. Rebuilt ASan/UBSan versions of the same two tests pass with `ASAN_OPTIONS=detect_leaks=0`; LeakSanitizer itself cannot run under this execution wrapper and fails before evaluating solver memory. The corrected candidate, not the crashing prototype, is the immutable binary measured by job `1645791`. It is correct but neutral and was fully reverted after measurement. No crash artifact, staged CNF, proof, or checker derivative remains.

4 Decision and next actions

No solver behavior or default changes in this increment. The accepted branch still contains the reserved-trail representation and all earlier Sweep, directed-bridge, BVE-admission, proof, and N1–N4 closures. The six new campaigns are reopening barriers: they require contradictory owner-local and whole-prefix evidence, not merely another timing sample.

The next source comparison should price Tactix's per-literal 16-byte `WatchHead` plus flat-arena offset arithmetic against Kissat's direct begin/end vector representation. First measure scan-entry indirection and the population/cost of delayed relocations without changing watch order. Any candidate must again preserve exact logical ticks/costs, candidate order, scheduling order, learning and pruning at common state. A material winner then requires exact event-trace parity, full Release and sanitizer coverage, the complete frozen B28 plus predeclared C5 guard with checked models/proofs and zero residue, exact-source fuzzing, and a checked two-CPU Rank-27 witness. Frozen reference solvers remain untouched.